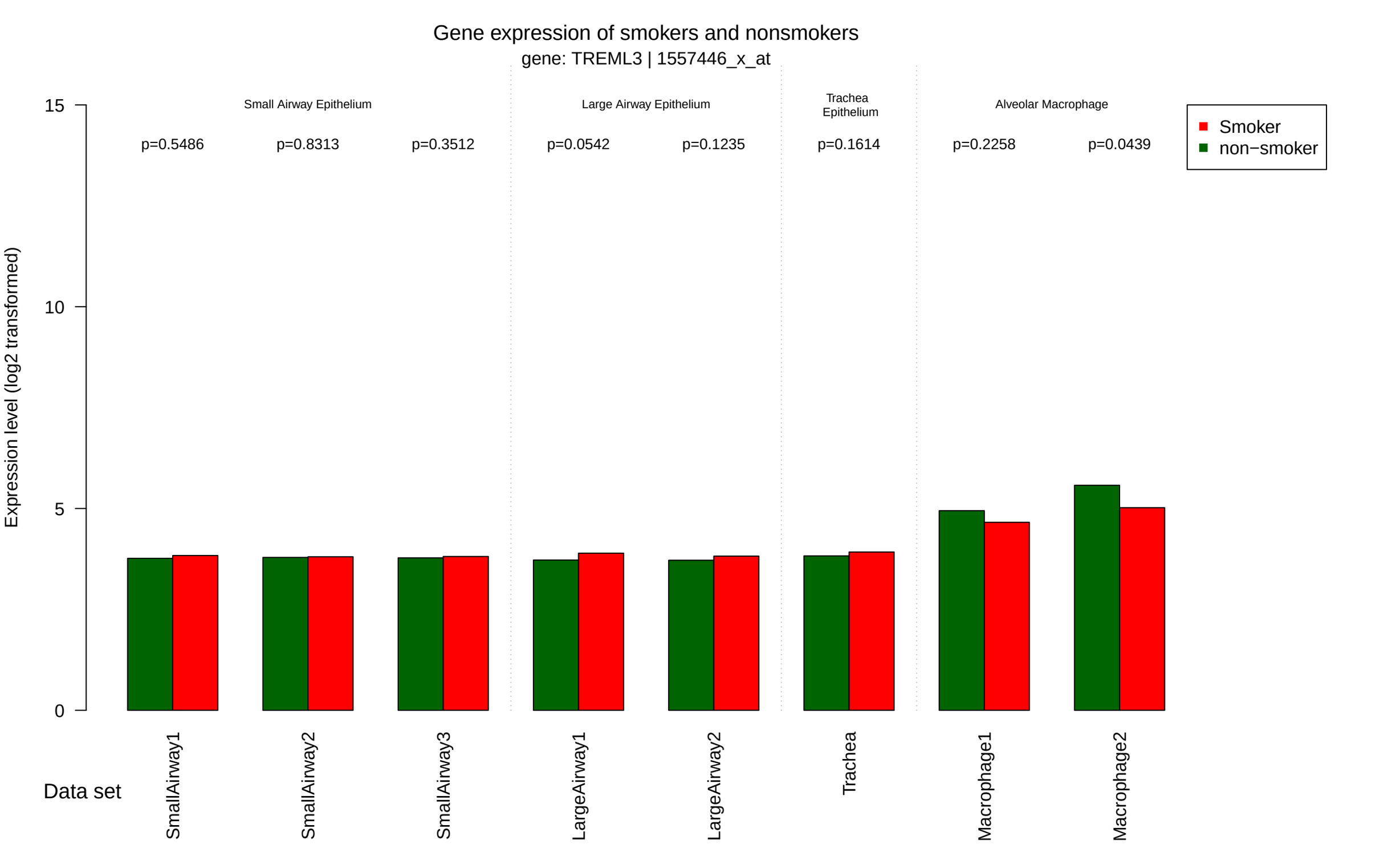
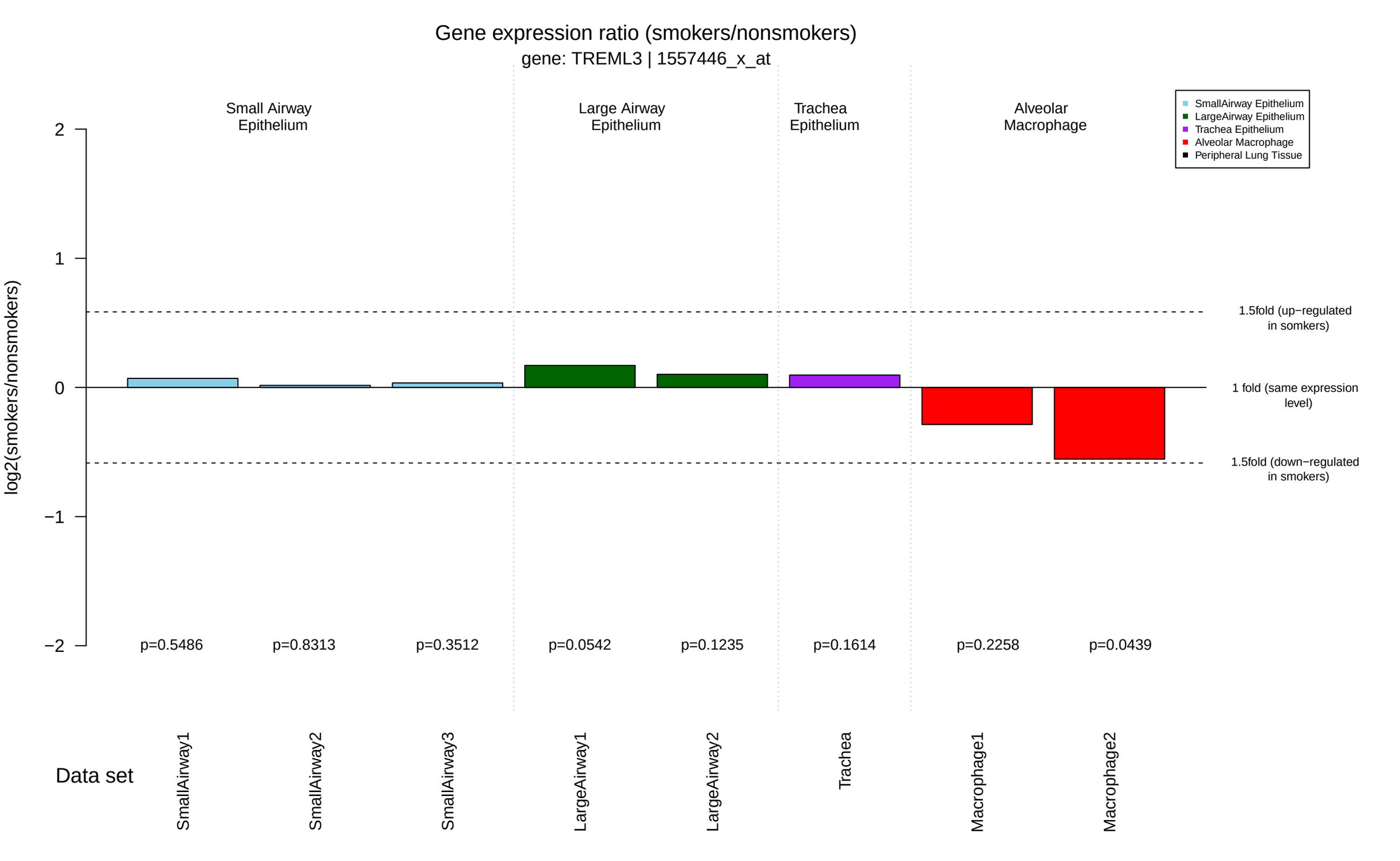
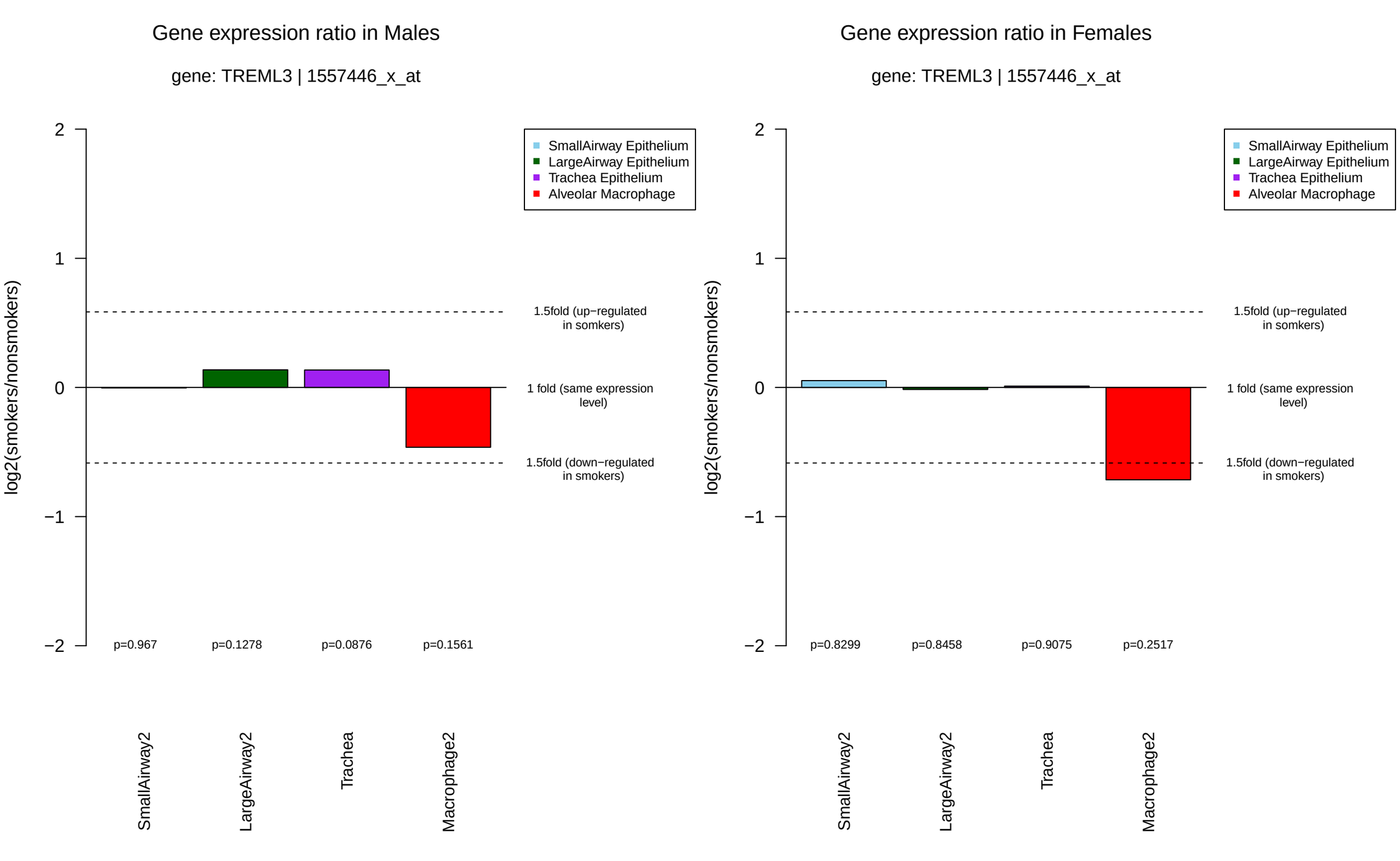
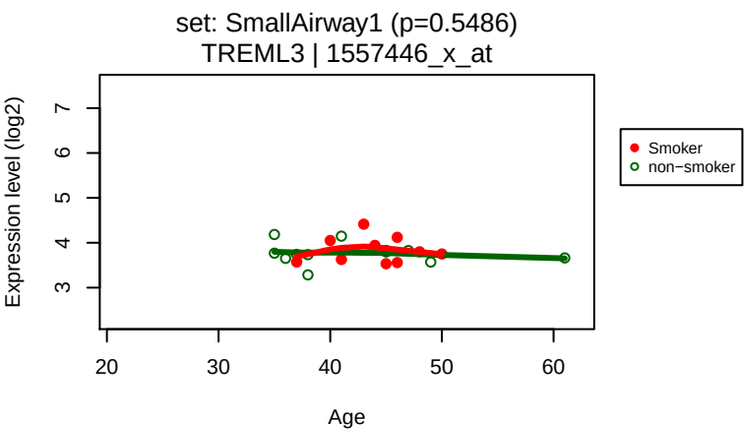
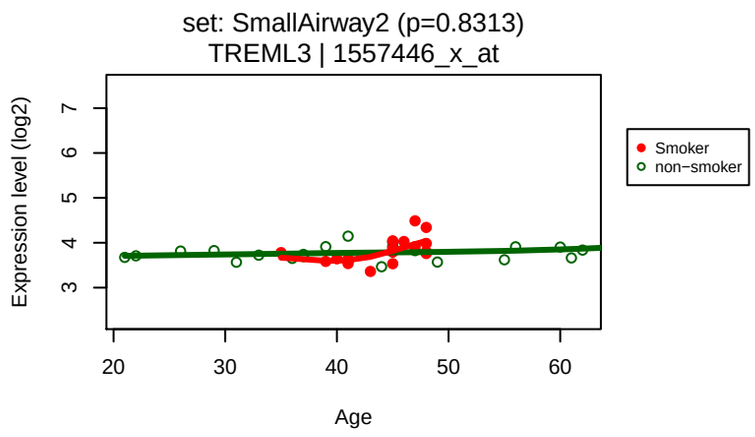
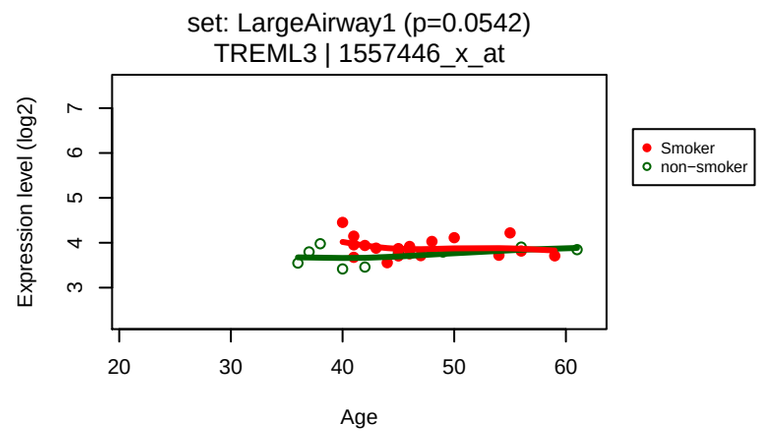

Gene expression of smokers and nonsmokers
gene: TREML3 | 1557446_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5486
p=0.8313
p=0.3512
p=0.0542
p=0.1235
p=0.1614
p=0.2258
p=0.0439
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

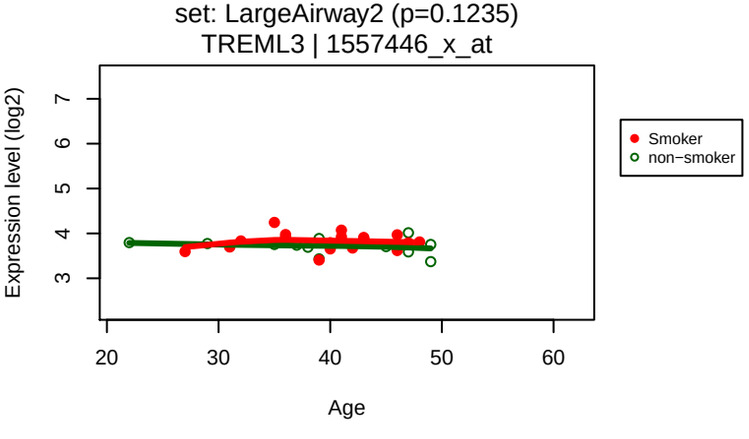

Gene expression ratio (smokers/nonsmokers)
gene: TREML3 | 1557446_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5486
p=0.8313
p=0.3512
p=0.0542
p=0.1235
p=0.1614
p=0.2258
p=0.0439
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

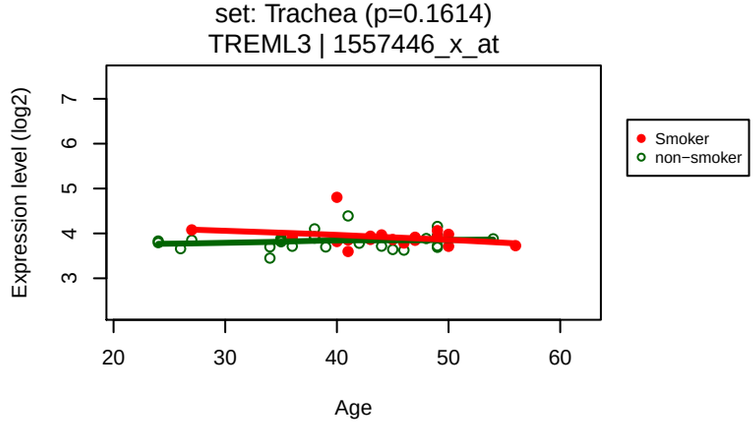

Gene expression ratio in Males
gene: TREML3 | 1557446_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.967
p=0.1278
p=0.0876
p=0.1561
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TREML3 | 1557446_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8299
p=0.8458
p=0.9075
p=0.2517
SmallAirway2
LargeAirway2
Trachea
Macrophage2

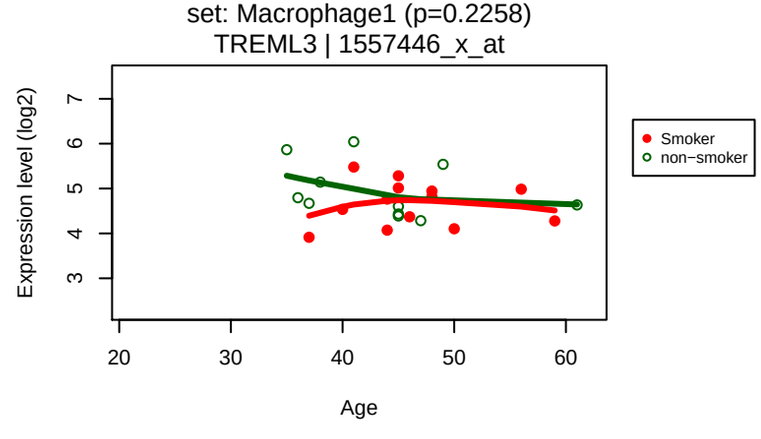

set: SmallAirway1 (p=0.5486)
TREML3 | 1557446_x_at
Expression level (log2)
Age
Smoker
non−smoker

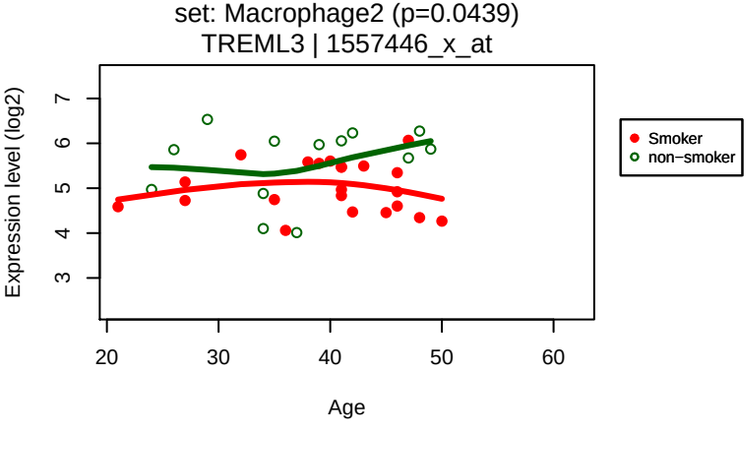

set: SmallAirway2 (p=0.8313)
TREML3 | 1557446_x_at
Expression level (log2)
Age
Smoker
non−smoker

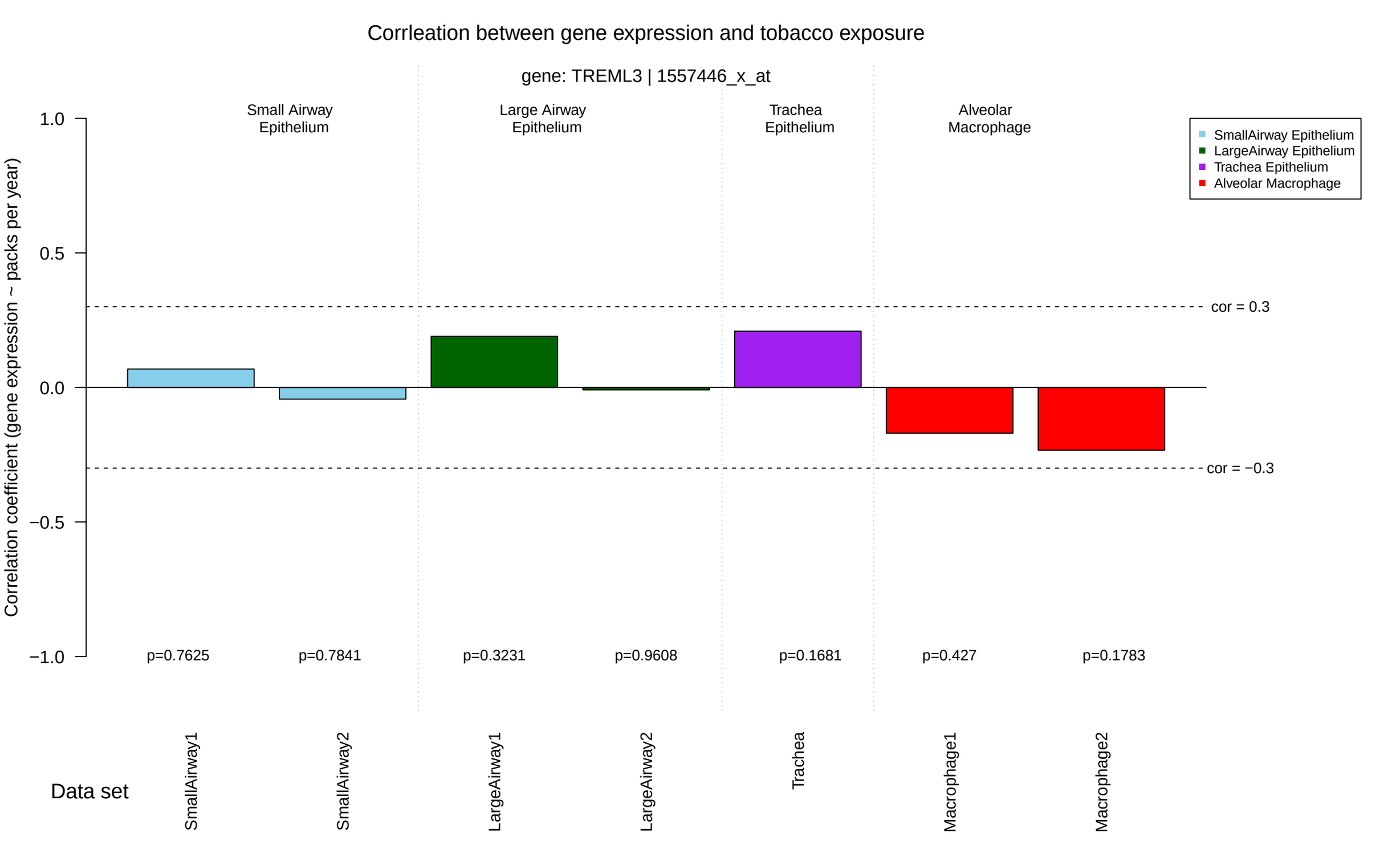

set: LargeAirway1 (p=0.0542)
TREML3 | 1557446_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1235)
TREML3 | 1557446_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1614)
TREML3 | 1557446_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2258)
TREML3 | 1557446_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0439)
TREML3 | 1557446_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TREML3 | 1557446_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7625
p=0.7841
p=0.3231
p=0.9608
p=0.1681
p=0.427
p=0.1783
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2